

Homework 1

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Problem 1.5

When asked to state the simple linear regression model, a student wrote it as follows:

$$E\{Y_i\} = \beta_0 + \beta_1 X_i + \epsilon_i$$

Do you agree?

Answer

No. The thing that ruins their answer is that it is giving the expected value of Y_i . That changes things. The book on page nine points out that:

$$E\{Y_i\} = \beta_0 + \beta_1 X_i$$

Note the lack of an epsilon at the end.

1.7

In a simulation exercise, regression model applies with $\beta_0 = 100$, $\beta_1 = 20$ and $\sigma^2 = 25$. An observation on Y will be made for $X = 5$.

Part A

Can you state the exact probability that Y will fall between 195 and 205? Explain?

No. Not unless I know the distribution of the error is normal.

Part B

Yes.

$$E(\hat{Y}) = \beta_0 + \beta_1 X_i = 100 + 20(5) = 200$$
$$\sigma = \sqrt{25} = 5$$

```
bounds <- c(195,205)
mean <- 200
sd <- 5

# This creates a vector of two probabilities, one from (-inf,195)
# and another from (-inf, 205).
probs <- pnorm(bounds, mean, sd)

# Subtract upper bound prob from lower bound prob
# Equivalent to normalcdf(195,205,)
prob <- probs[2] - probs[1]
```

The probability the predicted value falls between (195, 205) is approximately 0.683

1.18

$\sum e_i = 0$ when regression model 1.1 is fitted to a set of n cases by the method of least squares. Is it also true that $\sum \epsilon_i = 0$?

I would assume that no, it doesn't apply. As ϵ is an unknown batch of random variables that are independent and uncorrelated so the sum of these values is also random and unknown.

1.19

```
data_pr19 <- read.table("CH01PR19.txt", header = T)
reg_19 <- lm(gpa ~ act.score, data=data_pr19)
reg_19 |> summary()
```

```

Call:
lm(formula = gpa ~ act.score, data = data_pr19)

Residuals:
    Min       1Q   Median       3Q      Max
-2.74004 -0.33827  0.04062  0.44064  1.22737

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.11405     0.32089   6.588 1.3e-09 ***
act.score    0.03883     0.01277   3.040 0.00292 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6231 on 118 degrees of freedom
Multiple R-squared:  0.07262,    Adjusted R-squared:  0.06476
F-statistic:  9.24 on 1 and 118 DF,  p-value: 0.002917

```

Part A

$$\begin{aligned}\beta_0 &= 2.114 \\ \beta_1 &= 0.038 \\ \hat{Y} &= 2.114 + 0.038X\end{aligned}$$

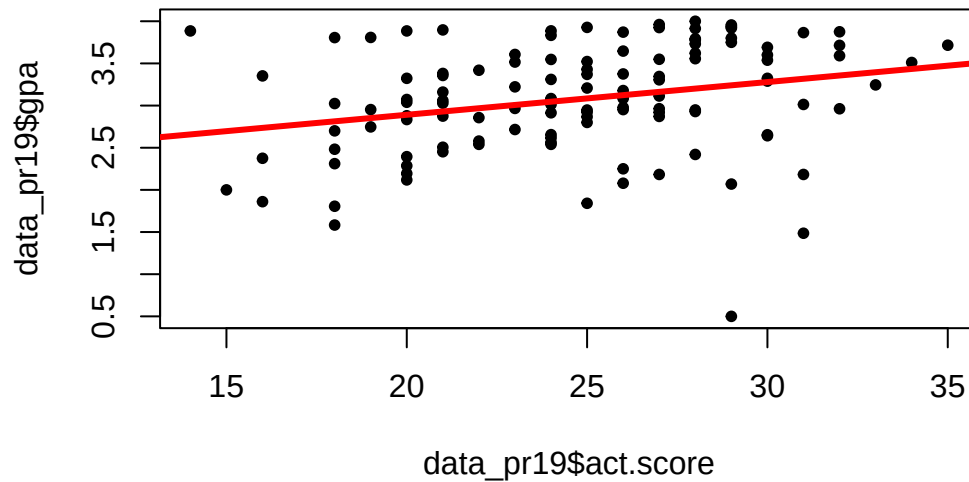
Part B

```

plot(data_pr19$act.score, data_pr19$gpa, main = "ACT Scores vs. GPA", pch=20)
abline(reg = reg_19, col="red", lwd = 3)

```

ACT Scores vs. GPA



This line doesn't appear to fit the data *well* and the R^2 is only 0.07 which indicates some large deviation away from the line. Regardless, there does seem to be a noticeable though slight positive relationship here.

Part C

```
predict(reg_19, newdata = data.frame(act.score=30))
```

```
1  
3.278863
```

Part D

```
predict(reg_19, newdata = data.frame(act.score=31))
```

```
1  
3.31769
```

1.21

```
data_pr21 = read.table("CH01PR21.txt", header = T)
reg_21 <- lm(ampules ~ transfers, data=data_pr21)
reg_21 |> summary()
```

Call:

```
lm(formula = ampules ~ transfers, data = data_pr21)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.2	-1.2	0.3	0.8	1.8

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.2000	0.6633	15.377	3.18e-07 ***
transfers	4.0000	0.4690	8.528	2.75e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.483 on 8 degrees of freedom

Multiple R-squared: 0.9009, Adjusted R-squared: 0.8885

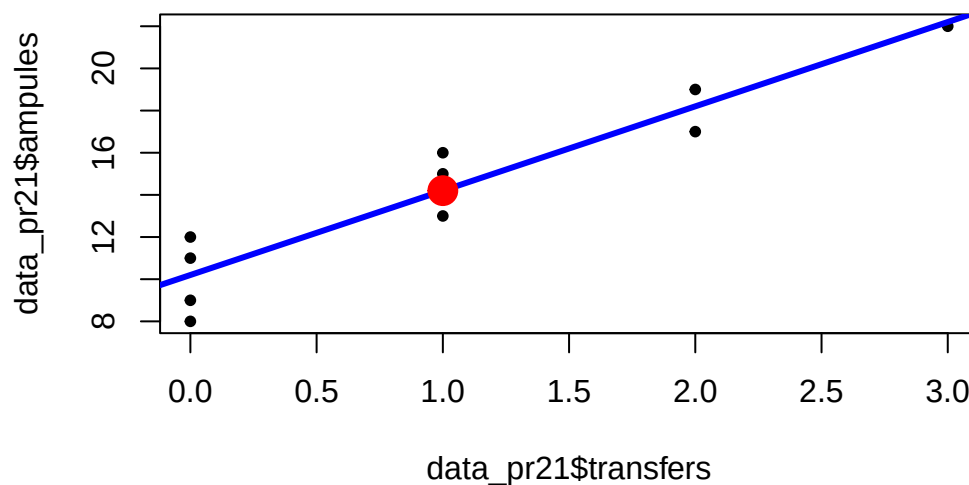
F-statistic: 72.73 on 1 and 8 DF, p-value: 2.749e-05

Part A and Part D

$$\hat{Y} = 10.2 + 4X$$

```
plot(x=data_pr21$transfers, y=data_pr21$ampules, pch=20, main="Transfers vs. Ampules")
abline(reg_21, col="blue", lwd=3)
points(x=mean(data_pr21$transfers), y=mean(data_pr21$ampules), col = "red", pch=20, cex=3)
```

Transfers vs. Ampules



The line appears to be a good fit with a high R^2 of 0.9.

Part B

```
new_data = data.frame(transfers = 1)
predict(reg_21, new_data)
```

1
14.2

Part C

Going from 1 transfer to 2 transfers it is expected for the number of broken ampules to increase by 4.

Part D

The red dot on the plot above from part B represents $(\bar{X}, \bar{Y})$. As the line crosses through the line this has been verified.

1.23

Part A

```
reg_19$residuals |> sum()
```

```
[1] -4.662937e-15
```

This only doesn't show 0 due to a computational rounding error. This is essentially 0.

Part B

```
sd_reg <- reg_19$residuals |> sd()  
var_reg <- sd_reg^2  
c(sd_reg, var_reg)
```

```
[1] 0.6205013 0.3850219
```

σ is expressed in units of the response variable, which in this case is destroyed ampules.

1.25

Part A

```
reg_21$residuals[1]
```

```
1  
1.8
```

1.8 is the first residual but after reading what I could I still really do not understand the relationship between this and the first random error ϵ_1 .

Part B

```
reg_21$residuals^2 |> sum()
```

```
[1] 17.6
```

```
reg_21 |> anova()
```

Analysis of Variance Table

Response: ampules

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
transfers	1	160.0	160.0	72.727	2.749e-05 ***
Residuals	8	17.6	2.2		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$$\sum e_i^2 = 17.6$$
$$MSE = 2.2$$

MSE estimates the variance of the residuals.